

2016.0 RANGE ROVER (LG), 419-07

NAVIGATION SYSTEM

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Navigation System, refer to the relevant Description and Operation section in the workshop manual. REFER to: [Navigation System](#) (415-01 Information and Entertainment System, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

ELECTRICAL

- Fuses
- Wiring harnesses and connectors
- Navigation control module
- Touch screen
- Audio amplifier module
- Speakers

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes

(DTCs) and refer to the DTC Index

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Navigation soft key missing from Touch Screen (TS) menu	▪ Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
"Please insert map SD card to enter navigation" message after launching navigation	▪ Map SD card missing	▪ Insert map SD card into the navigation control module
"Agree" message after launching navigation	▪ Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as

		necessary. Clear the DTCs and retest
Blank screen after launching navigation	- Video in signal missing - Navigation Control Module (NCM) not installed - LVDS circuit short circuit to ground, short circuit to power, open circuit, high resistance - Navigation Control Module (NCM) power or ground circuit open circuit, high resistance - Medium speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	GO to Pinpoint Test A.
Unable to detect position	- GPS signal missing - GPS receiver circuit short circuit to ground, short circuit to power, open circuit, high resistance	GO to Pinpoint Test B.
No navigation audio	- Telephone mute circuit short circuit to ground - Navigation audio circuits short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and test the telephone mute circuit for short circuit to ground - Refer to the electrical circuit diagrams and test the audio circuits for short circuit to ground, short circuit to power, open circuit, high resistance

PINPOINT TESTS

NOTES:

- The navigation control module **is** installed at the factory for China, Hong Kong, Taiwan and Macau market vehicles.
- The navigation control module **is not** installed at the factory for Korea, India and Israel market vehicles.

PINPOINT TEST A : NAVIGATION CONTROL MODULE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: CONFIRM INSTALLATION	
	1 Verify if the Navigation Control Module (NCM) has been installed.
	Has the Navigation Control Module (NCM) been installed (refer to the Navigation Control Module Locations table)? Yes GO to A2. No If applicable to market, install a new navigation control module.
A2: NAVIGATION CONTROL MODULE POWER AND GROUND TEST	
	1 Verify that the red LED on the Navigation Control Module (NCM) is illuminated (adjacent to the round connector).
	Is the red LED illuminated? Yes Refer to the electrical circuit diagrams and test the LVDS circuit to the Touch Screen (TS) for short circuit to ground, short circuit to power, open circuit, high resistance No GO to A3.
A3: NAVIGATION CONTROL MODULE POWER/GROUND TEST	
	1 Refer to the electrical circuit diagrams and test the Navigation Control Module (NCM) power and ground circuits for short circuit to ground, short circuit to power, open circuit, high resistance.
	Are the Navigation Control Module (NCM) power and ground circuits within specification? Yes GO to A4. No Repair the power or ground circuit as necessary

A4: MEDIUM SPEED CAN TEST

	1 Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the medium speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
	Is the medium speed CAN bus circuit within specification? Yes Repair the medium speed CAN bus circuit as necessary No Install a new navigation control module

PINPOINT TEST B : GPS TEST

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: GPS SIGNAL TEST	
	1 Remove the map SD card from the navigation control module
	2 Launch navigation by operating the navigation soft key on the Touch screen (TS)
	Is "GPS open" or "GPS short" displayed in a red box? Yes Refer to the electrical circuit diagrams and check the GPS receiver circuit to short circuit to ground, short circuit to power, open circuit, high resistance No "GPS open" and "GPS short" displayed in green boxes indicates that the GPS signal is correct

NAVIGATION CONTROL MODULE INSTALLED LOCATIONS

MODEL	LOCATION
XF - X250	Under the front left seat
XJ - X351	In the luggage compartment behind the left side trim panel
XK - X150	Navigation system not fitted
F-Type - X152	In the luggage compartment under the load floor at the right side
Freelander 2 - L359	Under the front right seat

Discovery 4 - L319	Under the front left seat
Evoque - L538	In the luggage compartment under the load floor at the right side
Range Rover - L405	In the luggage compartment under the load floor at the left side
Range Rover Sport - L320	Under the front left seat
Range Rover Sport - L494	In the luggage compartment under the load floor at the left side

D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

[Diagnostic Trouble Code Index - DTC: Navigation Control Module \(NCM\) - Extended Markets](#) (Description and Operation),
[Diagnostic Trouble Code Index - DTC: Navigation Control Module \(NCM\) - India & Israel](#) (Description and Operation).